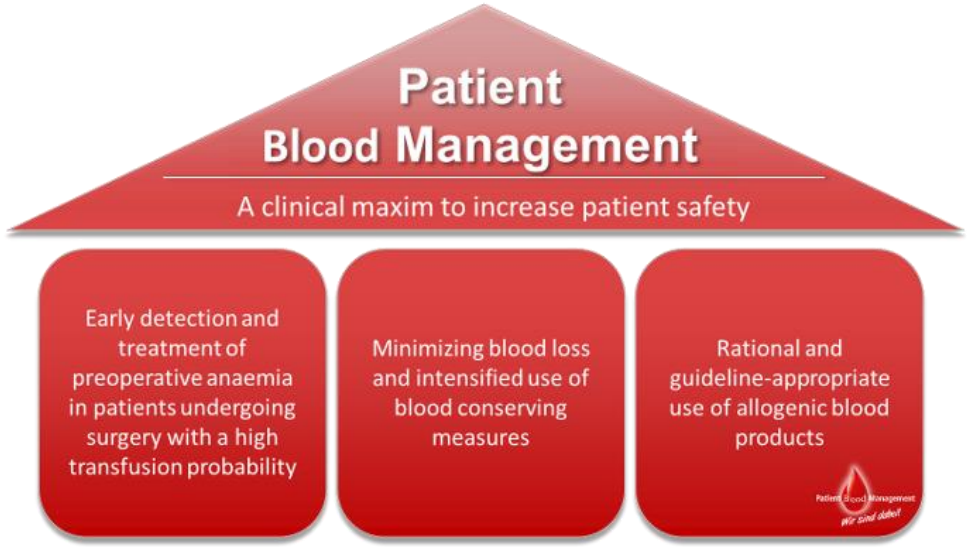
Cell salvage devices, centrifugal vs. filtration methods: Are we ready for clinical implementation?

Anouk Lindelauf, PhD
Maastricht UMC+
13-11-2025

Why is Patient Blood Management (PBM) important?

- Reduces transfusion-related morbidity and mortality.
- Optimises haemoglobin, minimises blood loss, and improves outcomes.
- Proven effective in cardiac surgery.

(Spahn DR et al., Anesthesiology 2019;130: 745–755)



What do the EACTS/EACTAIC 2024 guidelines Postoperative Drainage & Autotransfusion (Chapter 6.1) state:

- Controlled postoperative drainage, supported by digital systems, prevents haemothorax, infection, and delayed recovery while reducing complications and tube duration.
- Postoperative collected shed blood contains viable cells that, when clean and controlled, can be safely re-infused.
- This postoperative cell-salvage approach may support recovery and reduce allogeneic transfusion in bleeding patients.

What is the reason the use is still limited compared to Cardiac surgeries:

- Highest blood loss and transfusion rates → clear benefit.
- Postoperative anaemia is a common problem in the ICU. (Schmidbauer SL, et al., J Clin Med. 2022;11(13):3848)
- Implementation barriers: Conventional Cell Saver systems are not practical in the ICU or post-operative setting. (Kozek-Langenecker SA et al., Br J Anaesth 2023; 131: 993–1004)

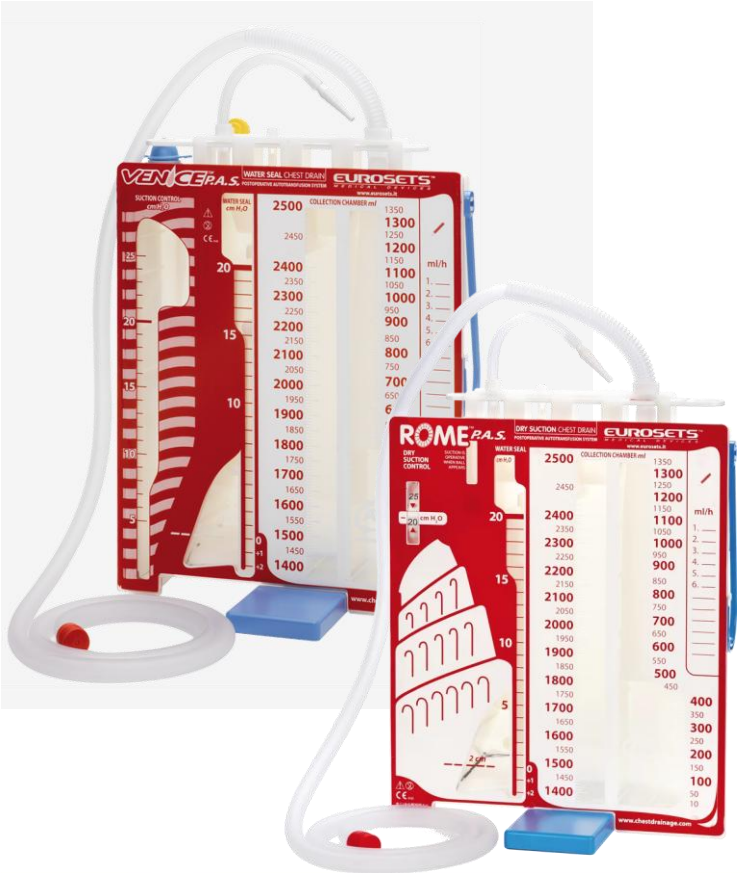


Why HemoClear® :

- Aligns with PBM, EACTS/ EACTAIC, goals: reduces allogeneic transfusion, preserves patient's own cells, and supports faster recovery.
- Affordable and bedside-applicable, ideal for non-cardiac and (cardiac) ICUs
- Enables clean and controlled reinfusion of postoperative shed blood.
- Preserves platelets unlike cell savers that remove this vital component.
- Simple, gravity-based system: no pumps, electricity, or complex setup.

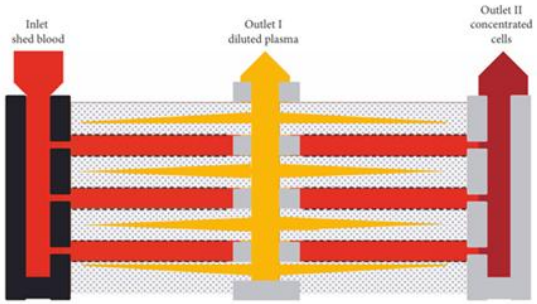
Study design post-op HemoClear cell salvage evaluation:

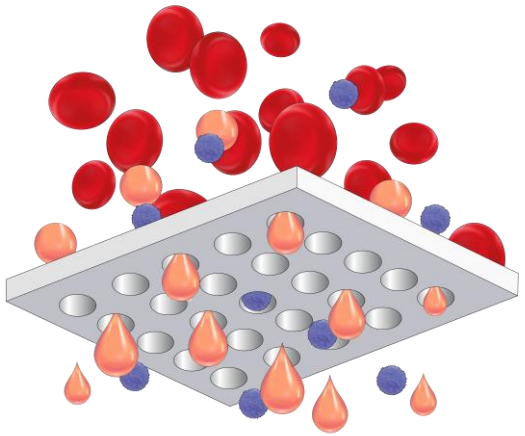
- Population: Adult cardiac surgery patients with ≥100 mL postoperative drainage.
- Collection: Sterile sampling of shed mediastinal blood within 6 h post-surgery, using Eurosets Venice Pass.
- Method: Compare unfiltered vs. HemoClear-filtered samples.
  - Haemoglobin
  - Haematocrit
  - Platelet count
  - Free Haemoglobin
  - D-Dimers

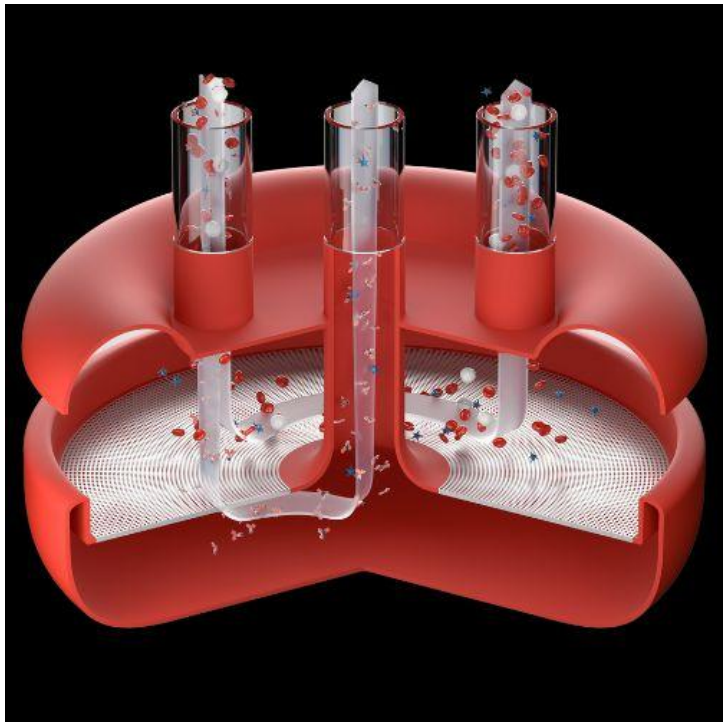


Processing technology:









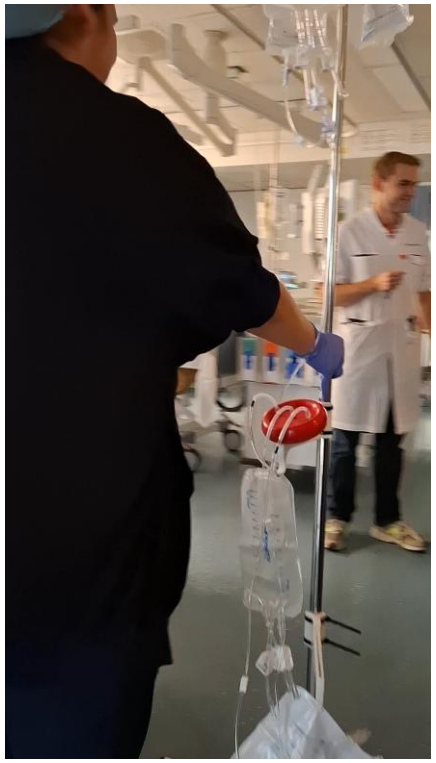
Study design, set-up:



1. Collection



Transfer line from drain outlet to filtration system blood bag



2. Process



3. Analyse

Study design, set-up: Processing protocol

- ***Figure a:*** blood from collection reservoir to HemoClear bag (1).
- ***Figure b:*** blood from transfusion bag (1) is being processed through the HemoClear together with NaCl to wash the blood and is collected in a second transfusion bag (2).
- ***Figure c:*** To start the second filtration process, transfusion bag 1 and 2 have changed in position. This filtration process will result in the final product (PRBC) in the first transfusion bag (1). In some cases however, a third filtering process is needed to create the final product.



(a)

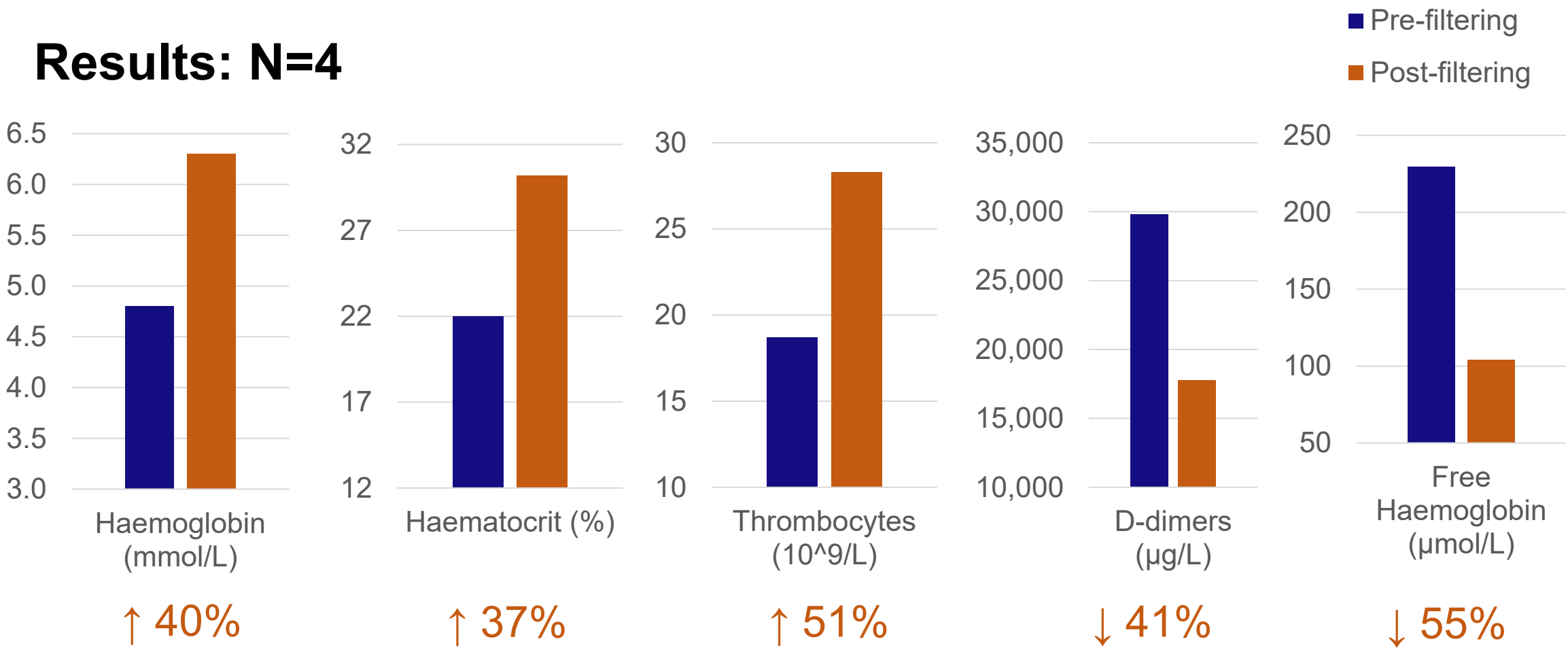


(b)



(c)

Results: N=4


Pre-filtering
Post-filtering
6.5
6.0
5.5
5.0
4.5
4.0
3.5
3.0
Haemoglobin (mmol/L)
↑ 40%
32
27
22
17
12
Haematocrit (%)
↑ 37%
30
25
20
15
10
Thrombocytes (10^9/L)
↑ 51%
35,000
30,000
25,000
20,000
15,000
10,000
D-dimers (µg/L)
↓ 41%
250
200
150
100
50
Free Haemoglobin (µmol/L)
↓ 55%

Implementation, procedures and training:

- Clinical protocol: define inclusion criteria, drainage volume threshold, and reinfusion timing.
- Sterile setup: integrate HemoClear® with chest drainage systems.
- Training: brief nursing and perfusion staff on collection, filtration, and reinfusion steps.
- Quality control: bedside visual check
- Documentation: record patient data, filtered volume, and reinfusion parameters.
- Governance: approval under hospital PBM or transfusion committee framework.
- Supply logistics: availability of single-use HemoClear® sets and sterile tubing.

Conclusion:

- Evaluation confirms that HemoClear® effectively bridges the postoperative cardiac PBM gap.
- It delivers a unique combination of clinical performance, efficiency, and operational simplicity.

Cell salvage devices, centrifugal vs. filtration methods: Are we ready for clinical implementation?

Anouk Lindelauf, PhD
Maastricht UMC+
13-11-2025